

PROBABILITY THEORY AND RANDOM PROCESSES (MA225)

LECTURE SLIDES

Lecture 13

Functions of Random Variables: Technique 1

Example 1: Let X_1 and X_2 be *i.i.d.* $U(0, 1)$ random variables. Find the CDF of $Y = X_1 + X_2$.

Example 2: Let the JPDP of (X_1, X_2) be given by

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} e^{-x_1} & \text{if } 0 < x_1 < x_2 < \infty \\ 0 & \text{otherwise.} \end{cases}$$

Find the JCDF of $Y_1 = X_1 + X_2$ and $Y_2 = X_2 - X_1$.

Functions of RVs: Technique 2 for DRV

Theorem: Let $\mathbf{X} = (X_1, X_2, \dots, X_n)$ be a DRV with JPMF $f_{\mathbf{X}}$ and support $S_{\mathbf{X}}$. Let $g_i : \mathbb{R}^n \rightarrow \mathbb{R}$ for all $i = 1, 2, \dots, k$. Let $Y_i = g_i(\mathbf{X})$ for $i = 1, 2, \dots, k$. Then $\mathbf{Y} = (Y_1, \dots, Y_k)$ is a DRV with JPMF

$$f_{\mathbf{Y}}(y_1, \dots, y_k) = \begin{cases} \sum_{\mathbf{x} \in A_{\mathbf{y}}} f_{\mathbf{X}}(\mathbf{x}) & \text{if } (y_1, \dots, y_k) \in S_{\mathbf{Y}} \\ 0 & \text{otherwise,} \end{cases}$$

where $A_{\mathbf{y}} = \{\mathbf{x} \in S_{\mathbf{X}} : g_i(\mathbf{x}) = y_i, i = 1, \dots, k\}$ and $S_{\mathbf{Y}} = \{(g_1(\mathbf{x}), \dots, g_k(\mathbf{x})) : \mathbf{x} \in S_{\mathbf{X}}\}$.

Functions of RVs: Technique 2 for DRV

Example 3: $X_1 \sim P(\lambda_1)$ and $X_2 \sim P(\lambda_2)$ and they are independent. Then $X_1 + X_2 \sim P(\lambda_1 + \lambda_2)$.

Example 4: $X_1 \sim \text{Bin}(n_1, p)$ and $X_2 \sim \text{Bin}(n_2, p)$ and they are independent. Then $X_1 + X_2 \sim \text{Bin}(n_1 + n_2, p)$.

Example 5: $X_i \sim \text{Bin}(n_i, p)$, $i = 1, 2, \dots, m$ and X_i 's are independent. Then $\sum_{i=1}^m X_i \sim \text{Bin}(\sum_{i=1}^m n_i, p)$.

Functions of RVs: Technique 2 for CRV

Theorem: Let $\mathbf{X} = (X_1, \dots, X_n)$ be a CRV with JPDF $f_{\mathbf{X}}$.

- ① Let $y_i = g_i(\mathbf{x})$, $i = 1, 2, \dots, n$ be $\mathbb{R}^n \rightarrow \mathbb{R}$ functions such that $\mathbf{y} = \mathbf{g}(\mathbf{x})$ is one-to-one. That means that there exists the inverse transformation $x_i = h_i(\mathbf{y})$, $i = 1, 2, \dots, n$ defined on the range of the transformation.
- ② Assume that both the mapping and its' inverse are continuous.
- ③ Assume that partial derivatives $\frac{\partial x_i}{\partial y_j}$, $i = 1, 2, \dots, n$, $j = 1, 2, \dots, n$, exist and are continuous.
- ④ Assume that the Jacobian of the inverse transformation

$$J \doteq \det \left(\frac{\partial x_i}{\partial y_j} \right)_{i,j=1,2,\dots,n} \neq 0$$

on the range of the transformation.

Then $\mathbf{Y} = (g_1(\mathbf{X}), \dots, g_n(\mathbf{X}))$ is a CRV with JPDF

$$f_{\mathbf{Y}}(\mathbf{y}) = f_{\mathbf{X}}(h_1(\mathbf{y}), \dots, h_n(\mathbf{y}))|J|.$$